

Microbial processing of plant residues in the subsoil – The role of biopores

Callum C. Banfield^{a,*}, Johanna Pausch^b, Yakov Kuzyakov^{c,d}, Michaela A. Dippold^a

^a University of Goettingen, Biogeochemistry of Agricultural Ecosystems, Buesgenweg 2, 37077 Goettingen, Germany

^b University of Bayreuth, Agroecology, Universitätsstrasse 30, 95440 Bayreuth, Germany

^c University of Goettingen, Department of Soil Science of Temperate Ecosystems, Buesgenweg 2, 37077 Goettingen, Germany

^d University of Goettingen, Department of Agricultural Soil Science, Buesgenweg 2, 37077 Goettingen, Germany



ARTICLE INFO

Keywords:

Detritusphere

Drilosphere

Earthworm tagging

Source partitioning

ABSTRACT

Most subsoil carbon (C) turnover occurs in biopore hotspots such as root channels and earthworm burrows. Biopores allocate large C amounts into the subsoil, where a vast capacity for long-term C sequestration is predicted. We hypothesise that organic matter (OM) cycling in biopores depends on their origin.

Earthworm and root biopores were induced under field conditions and were sampled from the subsoil (45–75 and 75–105 cm) after two years of biopore formation. The effects of biopore formation on OM decomposition were studied by biomarkers: neutral sugars, cutin and suberin-derived lipids, lignin-derived phenols and free lipids. The degradation stage of OM was biopore type-specific but was only governed by the soil depth in root biopores. Degradation of OM increased from earthworm biopores to root biopores and bulk soil. Hemicelluloses (GM/AX ratio) were more strongly degraded than lignin side-chains (relative change from initial values). Two years of microbial processing during biopore formation increased the GM/AX ratio in earthworm biopores from 0.65 to 1.05 and in root biopores from 0.15 to 1.35 (both relative to source biomasses). Root biopores and bulk soil had the highest GM/AX ratios (1.2–1.3), hinting to rapid processing of plant residues and accumulation of microbial residues. The regular, frequent OM inputs by earthworms stimulated microbial growth and processing of mostly bioavailable OM and, thus, relatively enriched more persistent OM (e.g. lignin). Syringyl subunits of lignin underwent low (ratio changed from 0.35 to 0.55 relative to initial input) and vanillyl subunits underwent almost no processing in earthworm biopores indicating the preferential microbial utilisation of the easily available compounds frequently replenished by earthworm activity. After two years of decomposition of the root detritus, mainly structural plant material was enriched in root biopores. Short periods (6 months) of earthworm activity effectively recharged the highly processed OM in root biopores with fresh OM.

In total, deep-rooting catch crops and short-term earthworm activities promote C accumulation in the subsoil followed by biopore-specific microbial processing predominantly governed by the C input frequency. As root biopores are up to 40 times more common than earthworm biopores, they dominate the OM input into subsoils. Such C inputs create several years lasting hotspots for preferential root growth and nutrient mobilisation in the subsoil. We conclude that root- and earthworm-derived biopores are vertical pathways for plant C from the soil surface into the subsoil and for intensive processing of litter C and sequestration of microbial necromass.

1. Introduction

Steadily increasing carbon (C) dioxide concentrations in the atmosphere are driving global climate change (IPCC, 2014). Its prominent projected consequences such as rising global mean temperatures and more variable precipitation (Li et al., 2009; Pal et al., 2004) are deemed unfavourable by society and therefore are to be mitigated (Tompkins et al., 2010; Urry, 2015). Sequestering carbon dioxide from the atmosphere in soils is being discussed as a viable option for reducing carbon dioxide concentrations (Lal et al., 2015; Poeplau and Don, 2015). Soils

are the largest pool in the terrestrial C cycle (Scharlemann et al., 2014; Schimel, 1995) and are comparatively easy to manage in comparison to geological or marine pools. This holds particularly true for croplands as they are under management. Usually, mostly the topsoils are considered for plant nutrition in arable fields as the main C cycling and nutrient pools are found in the top part of the soil (Kautz et al., 2013). Recently, the subsoils, i.e. the soil below the ploughed horizon, have been pushed into the centre of attention of soil science. Subsoils have the capacity of storing large amounts of C when their mineral phases (clay, iron oxides) are not saturated (Kell, 2012). The large land cover of cropland

* Corresponding author.

E-mail address: cbanfie@uni-goettingen.de (C.C. Banfield).

<https://doi.org/10.1016/j.soilbio.2018.08.004>

Received 26 March 2018; Received in revised form 25 July 2018; Accepted 3 August 2018

Available online 09 August 2018

0038-0717/ © 2018 Elsevier Ltd. All rights reserved.

(FAOSTAT, 2017), their manageability and the potential C storage capacity of subsoils make them attractive as long-term C sinks (Dignac et al., 2017; Torres-Sallan et al., 2017). However, for successful C sequestration, the dynamics of C in subsoils need to be better understood first.

Carbon reaches the subsoil predominantly through leaching as dissolved organic carbon (Kalbitz and Kaiser, 2008; Kindler et al., 2011). It involves potentially strong microbial modifications to the leached C compounds (e.g. respiration) and depends on the water solubility of the allocated compounds (Kaiser and Kalbitz, 2012). Large amounts of C may be also brought into the subsoils by roots or soil fauna such as earthworms, i.e. in biopores (Kautz, 2015; Stürzaker et al., 1996). Compared to leaching, organic matter (OM) ends up faster in the subsoil (Munyakusi et al., 1994), but in defined vertical macropores. They feature not just elevated C stocks, but also strong and intense microbial activity relative to the non-biopore bulk soil (Hoang et al., 2016). Hence, they are considered as hotspots, i.e. small soil volumes with much higher microbial processing rates than the average soil conditions (Kuzayakov and Blagodatskaya, 2015). The bulk subsoil is characterised by low C stocks, high radiocarbon age and slow microbial cycling (Rumpel et al., 2002; Rumpel and Kögel-Knabner, 2011). Biopores could contribute to higher C stocks in the subsoil because with higher C inputs, more C may remain stabilised as soil organic matter (SOM). Conversely, more easily available C may cause priming of older, potentially stabilised SOM, which would cause a loss of C (Fontaine et al., 2007; Kuzayakov, 2010). Studies on microbial communities and their residues in biopores have hinted to different functions of biopores, i.e. nutrient cycling vs. C sequestration – depending on their genesis, i.e. root vs earthworm-derived biopores (Banfield et al., 2017). C stabilisation may vary mechanistically among different hotspot types. Earthworms may change the texture of the soil in their burrows through selective feeding (Zhang and Schrader, 1993) and OM is strongly physically and chemically modified during the gut passage (Cheshire and Griffiths, 1989; Thakuria et al., 2010). Depending on the biopore-specific C inputs, the resulting microbial necromass may further contribute to long-term C sequestration (Miltner et al., 2012; Six et al., 2006). If C turnover depends on the biopore type, this will have consequences for the total subsoil C turnover, which mainly takes place in hotspots (Kuzayakov and Blagodatskaya, 2015).

The chemical composition of OM, its sources and degradation can be assessed by biomarkers (Amelung et al., 2008; Simoneit, 2005). Biomarkers are organic molecules whose detection indicates presence of an organism, tissue, secreted metabolites or their past presence. For instance, root influence is reflected by the contents of mid-chain ω -hydroxy alkanic acids and lignin-derived phenols (Armas-Herrera et al., 2016; Spielvogel et al., 2008; Thevenot et al., 2010). The degradation stage of the OM is assessed by biomarker ratios, e.g. contents of oxidised/reduced lignin-derived phenols (Thevenot et al., 2010). Ideally, undecomposed source biomasses (e.g. crop shoots as earthworm food) are collected before the experiment and characterised for their biomarker composition together with the soil samples (Gunina and Kuzayakov, 2015). The change of the biomarker pattern of the undecomposed source biomass to the biomarker pattern in soil can be related to the experiment duration, herein termed ‘microbial processing’.

To better understand C dynamics in hotspots and their relevance for subsoil C turnover, we conducted a biomarker study to link the biopore type and soil depth with C processing. Biomarker ratios known to characterise the degradation state of plant residues were used to reconstruct microbial processing by relating them to microbial biomarkers. In a field experiment over five years, three biopore types were induced by either (1) growing tap-rooted chicory (*Cichorium intybus* L.) for three years followed by two years of fallow (root biopores), (2) at least ≥ 3 years of earthworm activities (native earthworm biopores) or (3) 6 months of earthworm incubation into root biopores (earthworm-incubated biopores). We sampled the material on the inner walls of the

biopores and analysed it for lignin-derived phenols, cutin and suberin-derived biomarkers, neutral sugars and free lipids. Proxies describing the processing of plant residues were calculated from these data and compared among the biopore types.

2. Material and methods

2.1. Sampling

The study was conducted on the Campus Klein-Altendorf experimental research station near Bonn, Germany. The location had a mean annual temperature of 9.6 °C, a mean annual precipitation of 625 mm and featured a Haplic Luvisol (Hypereutric, Siltic; IUSS Working Group WRB, 2008). A characterisation of the soil and its genetic horizons was given in Vetterlein et al. (2013). The experimental set-up was previously described in Banfield et al. (2017). In short, three biopore types were induced in the field and studied in two subsoil depths (45–75 cm; 75–105 cm), namely root biopores, native earthworm biopores and a combination of both: earthworm-incubated biopores (Fig. S1, Supplementary Material). Each treatment combination was replicated four times in different locations of the same agricultural field.

- (I) *Root biopores*: from 2009 to 2012 chicory (*Cichorium intybus* L., var. Puna) was grown to induce taproots of at least 4 mm diameter in the subsoil. In 2012, the topsoil (0–45 cm) was temporarily removed and live roots were mapped in 45 cm depth. After filling the disturbed topsoil back, the plots were kept fallow and did not receive any clover-grass. Roots had two years to decay until sampling in autumn 2014.
- (II) *Earthworm-incubated biopores*: after 1.5 years of root decomposition, earthworms (*Lumbricus terrestris* L.), a native earthworm species in Central Europe, were inserted into a subset of at least 25 root biopores per field replicate for six months. In spring 2012, the topsoil (0–45 cm) was temporarily removed, and earthworms were incubated into previously identified and mapped 1.5-year-old root pores. Prior to incubation, an elastomer tag was injected into each earthworm body to allow re-discovery of the earthworms (Butt and Lowe, 2007). Earthworms were fed with clover-grass (shoots of *Trifolium repens* L., *T. pratense* L. and *Lolium perenne* L.) put on the soil surface until four weeks prior to sampling.
- (III) *Native earthworm biopores* (colonised with predominately *L. terrestris*) were treated similarly to the earthworm-incubated biopores: plots were kept fallow from 2012 on and grass-clover material was placed on the soil surface as food. Native earthworm biopores were identified at the end of the experiment in September 2014 as follows: after removing the topsoil, a new soil surface was prepared in –45 cm and covered with litter for three days. Biopores with visible earthworm middens were considered colonised with earthworms.
- (IV) *Bulk soil samples*, i.e. soil not containing any macroscopic biopores, were taken from areas next to the biopore plots. These plots were kept fallow and did not receive any clover grass amendment.

In September 2014, after removing the topsoil down to –45 cm, soil around each identified biopore was manually removed first down to –75 cm, then to –105 cm. The biopores were opened vertically and the inner wall coating was sampled by shaving it off with micro spatulas once, i.e. approximately 1 mm (Andriuzzi et al., 2013). Thirty-two samples were taken: four field replicates were taken from each treatment (three biopore types + bulk soil) and from two subsoil depths (45–75 cm; 75–105 cm). Each sample was pooled from about 25 biopores.

2.2. Biomarker analyses

Contents of *neutral sugars* from hemicelluloses and microbial

polysaccharides were determined after 4 M trifluoroacetic acid hydrolysis of 500 mg dried soil, followed by sample clean-up on DOWEX 50W X8 exchange resin and derivatisation to aldononitrile acetates after Zhang and Amelung (1996). Lignin-derived phenols were determined after alkaline CuO oxidation, sample purification (precipitation of humic acids and C18 solid phase extraction columns, SPE), derivatisation of the lignin-derived phenols to trimethylsilylates by N,O-Bis(trimethylsilyl)trifluoroacetamide. To separate *bound* and *free lipids*, 3 g of dried and ground soil were extracted in a Soxhlet apparatus for 36 h in a 2:1 (v:v) mixture of dichloromethane and methanol. The extracted soil was used for cutin/suberin analyses, while the solvent extract was used for the free lipid analysis. The extracted soil was hydrolysed in methanolic KOH to cleave monomers off the bound lipids *cutin* and *suberin*, cleaned up by liquid-liquid extraction at pH 1 and finally derivatized in two steps (methylation by BF₃: MeOH, 1.3M, and acetylation by acetic anhydride: pyridine). The *free lipids* consisted of the free *n*-alkanes, *n*-alkenes, *n*-alkanols + alkanones, *n*-alkanoic acids, α,ω -alkanedioic acids and ω -hydroxy alkanolic acids. Liquid-liquid extraction at two pH values separated the acidic from the neutral fraction, while silica gel SPE separated the substance classes. Carboxylic groups were methylated (cf. above), while hydroxyl groups were acetylated (cf. above). All analyses included internal standards to correct for analyte loss during preparation and for autosampler injection volume. Five to seven volumes of external standard solutions were derivatized with the samples of each biomarker approach and the regressions of the resulting peak areas and the concentrations of standards were used for analyte quantification. The samples were measured by gas chromatography on Agilent 7820A or 7890A GCs fitted with DB-5MS (lipids) or OV-17 (carbohydrates) columns coupled to mass selective detectors (Agilent 7000A) or flame ionisation detectors (all Agilent Technologies, Waldbronn, Germany). Full details about oven programmes, GC and MS parameters can be found in the Table S1 (Supplementary Material). The following proxies were calculated (Table 1):

- The ratio of microbial sugars to plant-derived sugars Σ (galactose + mannose)/ Σ (arabinose + xylose), in short: GM/AX, is a general proxy for the decomposition of plant material (Oades, 1984). With processing the proportion of plant-derived sugars decreases, whereas the proportion of microbial sugars successively increases.
- A source-adapted ratio of Σ (mannose + ribose + fucose)/ Σ (arabinose + xylose), in short MRF/AX, was calculated after

characterisation of the source biomasses. Into the biopores, e.g. root residues of chicory or shoot litter of clover grass was introduced. The source biomasses were analysed like the samples to receive their 'undecomposed' biomarker pattern. Substances were only used as plant biomarkers if they were identified in substantial amounts ($> 10\%$ of Σ) in the plant source biomasses such as chicory roots or clover shoots (Gunina and Kuzyakov, 2015; Moers et al., 1990; Oades, 1984; Rumpel et al., 2010). Hexoses were assumed to be microbial markers only if they were known in the literature (Tanaka et al., 1990) and if they were absent in the source biomasses (Gunina and Kuzyakov, 2015).

- The ratios of the lignin-derived syringyl/vanillyl phenols (S/V) or cinnamyl/vanillyl phenols (Ci/V) describe the lignin decomposition since syringyl and cinnamyl subunits are preferentially decomposed (Thevenot et al., 2010).
- The ratio of the acidic phenol to the aldehyde phenol for each lignin-derived subunit, e.g. (Ac/Al)_v for the vanillyl phenols, is a proxy for the side chain oxidation state of the lignin macromolecule (Thevenot et al., 2010).
- The ratio of Σ free/ Σ bound (ω -hydroxy alkanolic acids + α,ω -alkanedioic acids), i.e. solvent extractable / suberin-derived molecules, describes the breakdown of polymeric plant tissues (mostly root suberin), which are the quantitatively most relevant source for free hydroxy alkanolic acids $\geq$ C16 (Armas-Herrera et al., 2016; Bull et al., 2000; Kolattukudy, 1984), i.e. depolymerisation products (Naafs and van Bergen, 2002). Albeit minor sources include oxidative degradation of free alkanolic acids (Quenea et al., 2004), relating the contents of free to suberin-derived ω -hydroxy alkanolic acids was suggested to be a biomarker for suberin degradation (Otto and Simpson, 2007).
- Analogous to the above, the ratio of Σ free/ Σ bound saturated alkanolic acids was calculated as a potentially meaningful proxy for the cleavage of suberin. Longer alkanolic acids ($\geq$ C20) were suggested to be mainly derived from suberins (Bull et al., 2000; Kolattukudy, 1980; Naafs and van Bergen, 2002; Vidal et al., 2016).
- The ratio of Σ *n*-alkenes/ Σ *n*-alkanes $\geq$ C26 (both odd and even chain length) was calculated as a proxy for the microbial processing of cuticles and root tissues (Amblès et al., 1994; Vidal et al., 2016; Naafs and van Bergen, 2002). Long-chain *n*-alkanes derive from plant waxes (Lichtfouse, 1997; Wiesenberg et al., 2004), while *n*-alkenes are microbial products (Amblès et al., 1994; Lichtfouse et al., 1998).

Table 1

Overview of the biomarker proxies used to evaluate the microbial processing and degradation of plant inputs in biopores.

Biomarker	Abbreviation	Description	References
Σ (galactose + mannose)/ Σ (arabinose + xylose)	GM/AX	Processing of plant-derived hemicelluloses	(Gunina and Kuzyakov, 2015; Moers et al., 1990; Oades, 1984; Rumpel et al., 2010)
Σ (mannose + ribose + fucose)/ Σ (arabinose + xylose)	MRF/AX	Processing of plant-derived hemicelluloses, source-adapted	Source adaption (Gunina and Kuzyakov, 2015)
Ratio of syringyl/vanillyl lignin phenols	S/V	Processing of lignin-derived phenols (subunit-specific)	(Rasse et al., 2006; Thevenot et al., 2010)
Ratio of cinnamyl/vannilyl lignin phenols	Ci/V		(Rasse et al., 2006; Thevenot et al., 2010)
Ratio of acidic/aldehyde phenol for each lignin subunit	(Ac/Al) _v (vanillyl) (Ac/Al) _s (syringyl)	Oxidative degradation state of lignin-derived phenols	(Ertel and Hedges, 1984; Hedges and Ertel, 1982; Thevenot et al., 2010)
Ratio of Σ free/ Σ (bound ω -hydroxy alkanolic acids (C16–C24) + α,ω -alkanedioic acids (C16–C26))	HO-AA	Processing of suberin and root lipids	(Bull et al., 2000; Vidal et al., 2016)
Ratio of Σ free/ Σ bound <i>n</i> -alkanoic acids (C20–C30)	AA	Processing of suberin and root lipids	(Kolattukudy, 1980; Naafs and van Bergen, 2002)
Σ <i>n</i> -alkenes ($\geq$ C14)/ Σ <i>n</i> -alkanes ($\geq$ C26)	APR (alkene/precursor ratio)	Processing of epicuticular waxes ($\geq$ C26) to <i>n</i> -alkenes ($\geq$ C14)	(Amblès et al., 1994; Vidal et al., 2016)
Bulk isotopic composition	$\delta^{13}\text{C}$	General turnover marker for SOM	(Gunina and Kuzyakov, 2014; Werth and Kuzyakov, 2010)

- The C/N ratio and bulk isotopic composition ($\delta^{13}\text{C}$) are broad proxies for SOM processing (Balesdent et al., 1987).

These proxies needed to be related to the intact source biomasses since prior to decomposition the source biomasses already differ in their proxy values (Table S2, Supplementary material). We characterised the biomasses of *L. terrestris* bodies, *C. intybus* shoots, *C. intybus* roots, *Lolium perenne* shoots, *Trifolium pratense* shoots and *T. repens* shoots. Data on the bulk isotopic composition ($\delta^{13}\text{C}$), as well as C and nitrogen (N) contents of the same treatments, were taken from Banfield et al. (2017).

2.3. Statistical evaluation

Each dataset containing the proxy values was screened for outliers by Nalimov's test and maximum one outlier per treatment combination and proxy was removed. A principal component analysis (PCA) based on correlations was performed on the combined dataset of all proxies. For the treatments, factor coordinates were calculated from the two principal components with the highest eigenvalues. Likewise, for the variables (biomarker ratios), a correlation circle was calculated. The MRF/AX ratio was excluded due to collinearity with the GM/AX ratio. The grouping variable was the treatment (three biopore types and bulk soil). Data given in the tables and figures are mean values $\pm$ standard errors. Significant differences of means among biopore types were tested by one-way analyses of variance (ANOVA) for each depth separately since the two soil depths provide paired samples. Levene's test was used to test for homogenous variances. Normality of the residues was checked in Q-Q plots. Post-hoc comparisons were by Tukey's Honest Significant Differences Test. If assumptions were not met, non-parametric ANOVA (Kruskal-Wallis ANOVA) including post-hoc comparisons of mean ranks were used instead. Pairwise two-sample t-tests for dependent samples were used to determine differences between soil depths for each biopore type and proxy. All statistical analyses were performed in Statistica 13.3 (TIBCO Software Inc., Palo Alto, CA, U.S.A.).

3. Results

The PCA calculated from seven variables explained 51% of the total variance (Fig. 2). All variables, i.e. biomarker proxies, were positively correlated with the principal component (PC) 1 (x-axis, Fig. 2, right).

The GM/AX and (Ac/Al)v ratios were most strongly correlated with PC 1, which separated the earthworm biopores from root biopores and bulk soil (Fig. 2, left). Both earthworm biopore types showed little in-group variance and were overlapping. Earthworm incubation for six months obscured the former root presence by incorporation of fresh material. Bulk soil and root biopores showed much higher in-group variance along PC 2, which separated root biopore samples from to -45–75 cm and from of -75–105 cm. The effect of the soil depth was most strongly represented by the lignin biomarkers (S/V, (Ac/Al)s, (Ac/Al)v ratios; Fig. 2, right). Long-term and short-term earthworm activities in the earthworm biopores obscured any soil depth effect. The other proxies were equally correlated with both principal components. As the Ci/V ratio correlated only weakly with either axis, it was only included in the supplementary material (Fig. S2).

3.1. Neutral sugar ratios

The sugar ratios values did not differentiate the biopores types from the bulk soil in 45–75 cm but in 75–105 cm (Fig. 2). Earthworms decreased the degradation stage of the hemicelluloses in the deeper sub-soil (relative to bulk soil, i.e. 'earthworm effect' illustrated exemplarily in Fig. 2), while after two years of root decay in the root biopores the hemicelluloses were as degraded as in bulk soil. Relative to the source biomasses, native earthworm activity increased the GM/AX ratio from 0.65 (clover grass) to 1.0, while short-term earthworm incubation in root biopores increased it from 0.42 (50% clover grass + 50% roots) to 0.95. So, the means of the earthworm biopores were close to the source biomasses, suggesting that polysaccharides in earthworm biopores had undergone the least microbial processing among the biopores. In comparison, the GM/AX ratio after two years decomposition was 1.3 in root biopores, while the undecomposed root necromass had a GM/AX ratio of 0.2, which underlines the strong processing of polysaccharides compared to the earthworm biopores. The high GM/AX ratio of bulk soil reflected the long-term turnover. Equivalent results were found for the MRF/AX ratio.

3.2. Processing of the lignin macromolecule

The oxidation state of the lignin side chains (Fig. 3) confirmed the degree of OM processing in biopores previously seen in the polysaccharides – irrespective of the subunit studied (Fig. 2). Relative to

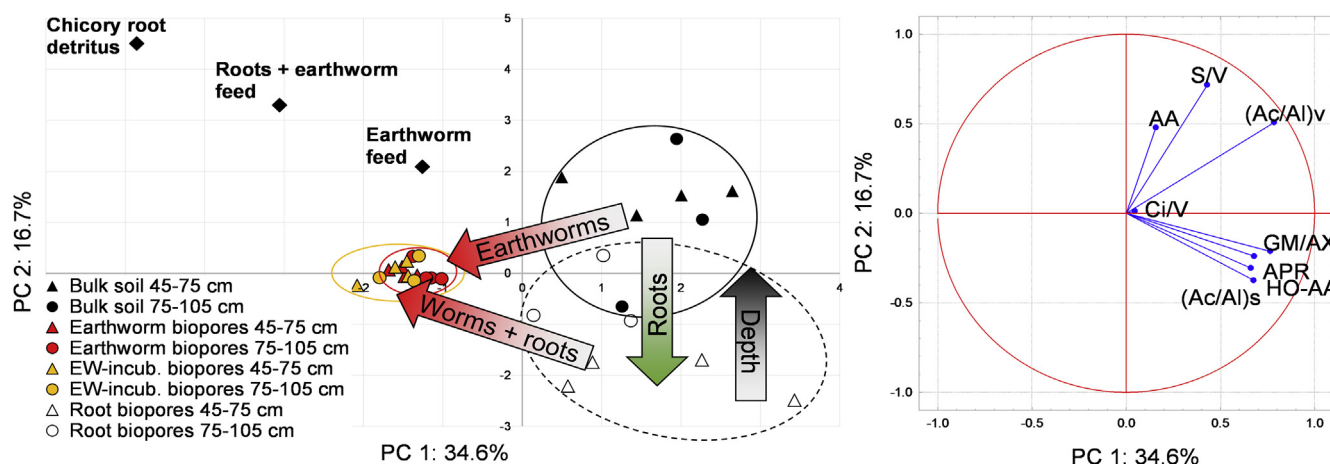
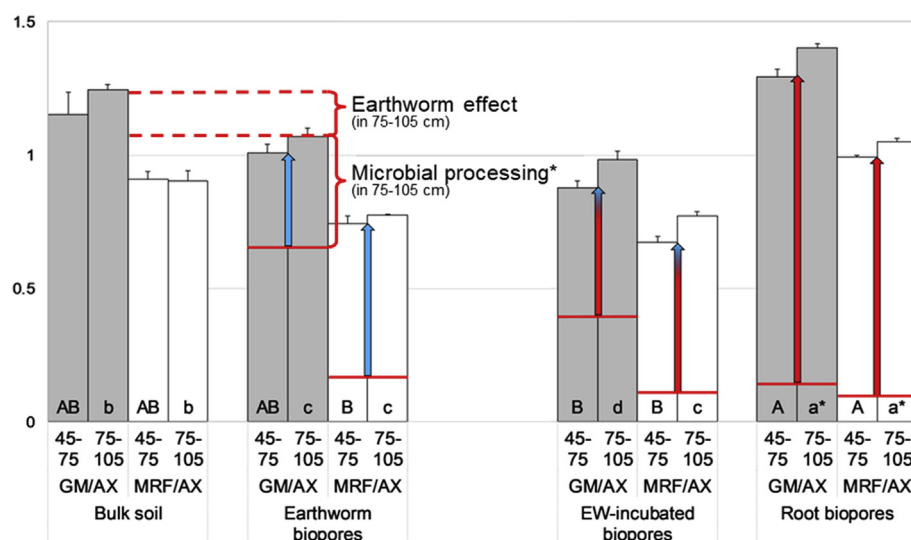


Fig. 1. Left: Principal component analysis of seven proxies describing the degree of processing: GM/AX, Ac/Al for S and V subunits, S/V and Ci/V ratios; free/bound ω -hydroxy alkanolic acids or alkanolic acids ratios, as well as the *n*-alkene/*n*-alkanes ratio. Source biomasses were added as supplementary data (black diamonds): earthworm feed, chicory root detritus and a combination of both. Different colours represent the four treatments (bulk soil vs native vs earthworm (EW)-incubated biopores vs root biopores). Triangles represent samples from 45–75 cm, while circles are samples from 75–105 cm. Ellipses show the within-group variance. PC 1 and 2 explained 35% and 17% of the inertia, respectively. The arrows illustrate the effects of earthworms (corresponding to PC 1, i.e. x-axis), soil depth (mainly corresponding to PC 2, i.e. y-axis), earthworm incubation and tap roots. **Right:** Correlation circle describing the correlation between biomarker ratios and the two PC. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)



red dashed line as a visual guide). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

bulk soil, the lignin oxidation state in earthworm biopores was lower in both subsoil depths. However, lignin in root-influenced biopores was only less oxidised in the 75–105 cm horizon. Highest oxidation was apparent in bulk soil. Both earthworm biopores had the lowest lignin oxidation state and also the least microbial processing during biopore formation (oxidation state increased from initial 0.4 → 0.55 for (Ac/Al) s). Vanillyl subunits underwent almost no processing in any of the earthworm biopore types. Like in the case of sugars, root biopore lignin underwent the strongest processing among biopores. The S/V ratio showed stronger processing in bulk soil than in earthworm biopores (Fig. S2, Supplementary material), especially in 45–75 cm. The Ci/V could only be applied to the earthworm biopore types since only in them cinnamyl subunits were recovered, which stem from the clover-grass feed. Preferential earthworm activities may be deduced from the Ci/V ratio since significant differences were only found in 45–75 cm between both earthworm biopore types.

3.3. Lipid-derived proxies

The HO-AA ratio, describing the depolymerisation of suberin to free ω -hydroxy alkanolic acids, was highest in bulk soil ($p < 0.05$ in 75–105 cm, Fig. 4). Suberin in earthworm biopores remained rather intact relative to bulk soil. The effects of root decay, earthworm incubation and earthworm burrowing on the depolymerisation state of suberin were similar in 75–105 cm, however, no data was available for 45–75 cm. Significant differences among biopores were found for the AA ratio, describing the depolymerisation of suberin to free alkanolic acids: highest values were found in the root biopores relative to native earthworm biopores (75–105 cm). The ratio of n -alkenes to their precursors n -alkanes (describing the degradation of epicuticular waxes) had a similar behaviour among treatments like the GM/AX ratio: the highest degree of processing in bulk soil and root biopores, lowest in both earthworm biopore types. However, both root detritus-influenced biopore types featured a lower degree of degradation of waxes with depth ($p < 0.05$). Summarising, the proxies for the hemicelluloses, suberin and lignin showed increasing OM processing in the order:

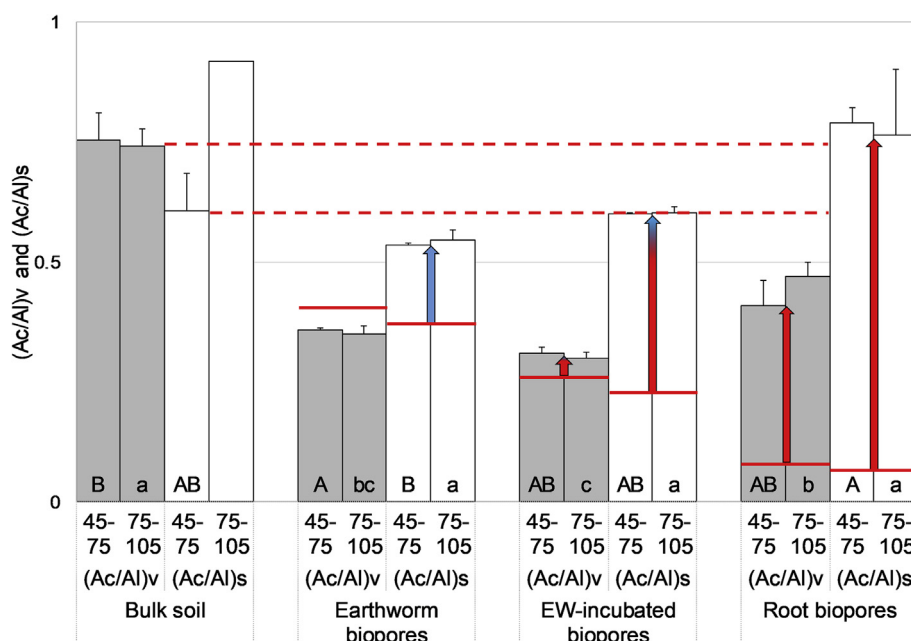


Fig. 3. Oxidation state of the lignin side chains. Ratios of vanillic acid/vanillin (Ac/Al)v as grey bars and syringic acid/syringaldehyde (Ac/Al)s as white bars. Shown are means $\pm$ standard errors of all four treatments. Capital letters indicate significant differences for 45–75 cm (one-way ANOVA, Tukey's HSD test on α 0.05), whereas small letters indicate significant differences for 75–105 cm. (Ac/Al)s for bulk soil represents only one value. Red lines indicate the values of the source biomasses, i.e. plant inputs prior to decomposition. The red and blue arrows indicate the effects of biopore-specific microbial processing. The difference to the bulk indicates the effects of roots and/or earthworms, respectively. The red dashed lines correspond to the bulk soil degradation state as a visual guide. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

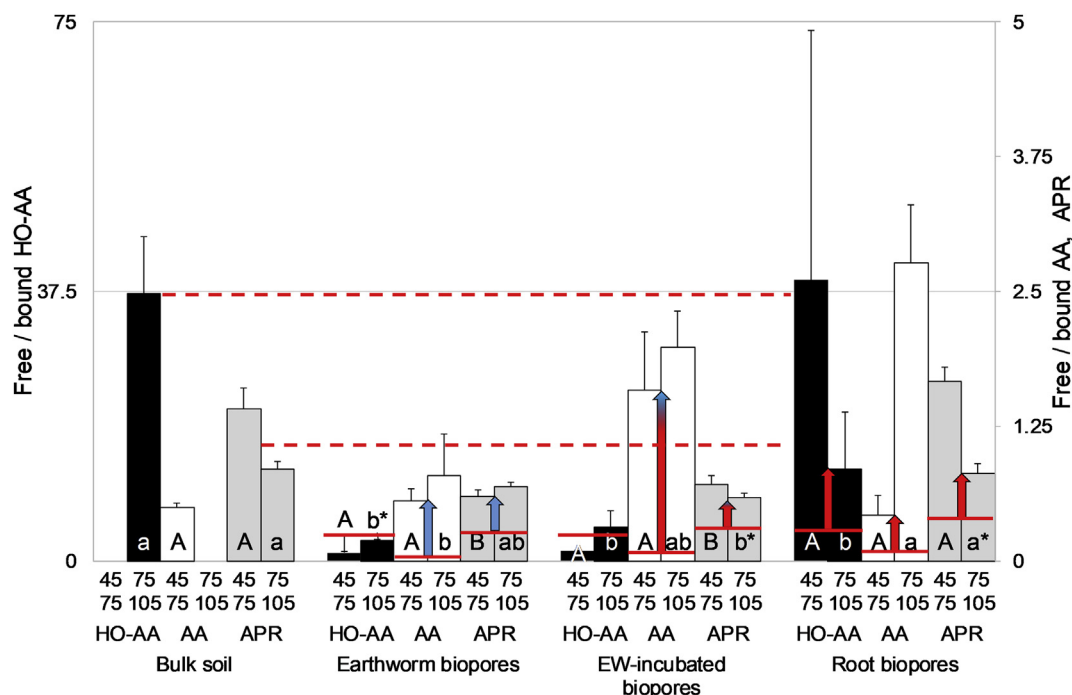


Fig. 4. Lipid-based proxies: free/bound ω -hydroxy alkanolic acids (HO-AA) as black bars on the left y-axis; free/bound alkanolic acids (AA) as white bars and Σ n -alkenes/ Σ n -alkanes > C26 as grey bars (APR), both on the right y-axis. Shown are means $\pm$ standard errors of all four treatments. Capital letters indicate significant differences for 45–75 cm (one-way ANOVA, Tukey's HSD test on α 0.05), whereas small letters indicate significant differences in 75–105 cm. Asterisks indicate significant differences of a variable between both soil depths (Two sample t -test for dependent samples). Red lines indicate the values of the source biomasses, i.e. plant inputs prior to decomposition. The red and blue arrows indicate the effects of biopore-specific microbial processing. The difference to the bulk indicates the effects of roots and/or earthworms, respectively. The red dashed lines correspond to the bulk soil degradation state as a visual guide. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

earthworm biopore < root biopores < bulk soil.

4. Discussion

4.1. Applicability of biomarker-derived proxies

This study employed biomarkers ratios to determine the degree of microbial processing of biopore OM and the effects of soil biota. Biomarker ratios have been used extensively in the past to study OM dynamics (Glaser et al., 2004; Otto and Simpson; Spielvogel et al., 2007). As with many biomarkers, their sources are frequently generalised and thus not always undoubted or specific (Amelung et al., 2008; Frostegård et al., 2011), e.g. in the case of neutral sugars (Gunina and Kuzyakov, 2015). Thus, we characterised the input biomasses (e.g. chicory roots, clover grass) to get source-adapted biomarkers. The lignin, suberin and hemicellulose markers represent the main structural components of shoot litter and root residues (Bull et al., 2000; Kögel-Knabner, 2002). They reflect medium and long-term decomposition processes and are not appropriate to characterise the very early stages of litter decomposition (Haider and Martin, 1979; Schöning et al., 2005).

As the time points of the last C input into different biopore types were not identical in our study, a direct comparison of the decomposition rates in the biopores is not quantitatively possible (Figs. 2–4). As under field management conditions inputs into different biopores also occur at different times of the cropping season (Eriksen-Hamel et al., 2009), our design features representative conditions for biopores in agricultural soils. Therefore, our findings can be generalised: the conditions described herein, i.e. earthworm burrowing and decaying roots, are common in arable fields under fallow or no-till (Chan, 2001). Certainly, the OM input amounts were not identical among the biopore types. By calculating biomarker ratios, the absolute biomarker contents are taken out of the equation and the decomposition stage under field

conditions becomes evident.

At a glance, all proxies present a certain degree of degradation of the biopore OM. The biopore development was intensive enough to affect all proxies irrespective of the analytical method and suggested mutual validation of the proxies (also visible in the correlation circle of Fig. 1, right). However, comparing e.g. lignin and n -alkane proxies in root biopores (Figs. 3 and 4), a depth effect was apparent only for the latter. Therefore, it is crucial to compare proxies for multiple substance classes to get the 'big picture', i.e. the compound and hotspot-specific OM transformations. Apart from the biomarker-derived proxies, the microbial processing also impacted the bulk isotopic composition ($\delta^{13}\text{C}$) and the C/N ratio through the relative increase of microbial necromass relative to the successively decomposed plant material (Table 2). As less pronounced treatment effects were apparent relative to the biomarkers, $\delta^{13}\text{C}$ was clearly a less sensitive marker for the processing of OM compared to the biomarker ratios.

The distinction between the GM/AX ratio and the source-adapted MRF/AX ratio was not immediately obvious in case of the investigated hotspots (Fig. 2). The similarity of both ratios underlines that both are suitable for C dynamics studies due to the same methodological background (Gunina and Kuzyakov, 2015). However, the MRF/AX ratio is less biased than the GM/AX ratio in our setup: a high galactose content in chicory roots and other plants (Angers and Mehuys, 1990), raises concerns if the frequently applied GM/AX ratio is always the best ratio. Galactose found in biopores of chicory may be mostly root-derived and not microbial. We, therefore, recommend checking the primary inputs for confounding sugar contents and calculating source-adapted biomarkers.

Reduced explanatory power may result from sampling: the bulk soil and the soil ingested by earthworms contains native lipids and carbohydrates (Gunina and Kuzyakov, 2015; Kögel-Knabner et al., 2008). This potentially influences the apparent degree of processing at the end of the experiment. In our setup, this effect is not expected to play a

Table 2

General soil properties and information regarding the biopores. Shown are mean values $\pm$ standard errors of TOC and TN contents, C/N ratios and $\delta^{13}\text{C}$ for root biopores, EW-incubated biopores, native earthworm biopores and bulk soil. Capital letters indicate significant differences for 45–75 cm (one-way ANOVA, Tukey's HSD test on α 0.05), whereas small letters correspond to 75–105 cm. Asterisks indicate significant differences of a variable between both soil depths (two sample *t*-test for dependent samples).

Data taken from Banfield et al. (2017).

	Bulk soil		Earthworm pores		EW-incubated pores		Root pores	
	45–75 cm	75–105 cm	45–75 cm	75–105 cm	45–75 cm	75–105 cm	45–75 cm	75–105 cm
Total organic carbon [%]	0.41 $\pm$ 0.02 ^A	0.35 $\pm$ 0.05 ^b	1.17 $\pm$ 0.05 ^C	1.05 $\pm$ 0.04 ^a	1.16 $\pm$ 0.04 ^C	1.07 $\pm$ 0.04 ^{**}	0.81 $\pm$ 0.03 ^B	0.93 $\pm$ 0.06 ^a
Total nitrogen [%]	0.06 $\pm$ 0.00 ^A	0.05 $\pm$ 0.00 ^{ba}	0.11 $\pm$ 0.00 ^C	0.10 $\pm$ 0.01 ^a	0.12 $\pm$ 0.00 ^C	0.11 $\pm$ 0.01 ^a	0.09 $\pm$ 0.00 ^B	0.10 $\pm$ 0.01 ^a
C/N	7.2 $\pm$ 0.1 ^D	7.7 $\pm$ 0.9 ^a	10.3 $\pm$ 0.2 ^C	10.3 $\pm$ 0.5 ^a	9.6 $\pm$ 0.1 ^B	9.9 $\pm$ 0.4 ^a	8.6 $\pm$ 0.2 ^A	9.7 $\pm$ 0.5 ^a
$\delta^{13}\text{C}$ [‰]	−25.0 $\pm$ 0.05 ^B	−23.5 $\pm$ 0.75 ^a	−25.3 $\pm$ 0.25 ^{AB}	−23.8 $\pm$ 0.66 ^a	−26.5 $\pm$ 0.21 ^A	−25.6 $\pm$ 0.51 ^a	−25.7 $\pm$ 0.46 ^{AB}	−23.4 $\pm$ 1.17 ^a

major role for two reasons. First, the lipid and carbohydrate contents in the deep bulk subsoil were rather low and the C inputs into the biopores were orders of magnitude larger resulting already after two years in a threefold higher C content (Banfield et al., 2017). Second, no or only little bulk soil was sampled with the biopore linings. Care was taken not to take all biopore linings by shaving off the inner walls only once with a micro spatula. Earthworm biopores are expected to be influenced by carbohydrate-containing mucus produced by the earthworms (Brown et al., 2000; Curry and Schmidt, 2007).

4.2. The effects of earthworms, roots and soil depth on the decomposition stage of biopore OM

The PCA illustrated the similarities between both earthworm biopore types and between root biopores and the bulk soil (Fig. 1, left): earthworm-influenced biopore types were fully overlapping with each other, and the root biopores were partly overlapping with the bulk soil. Earthworm activities were discernible in the PCA from smaller within-group variance than the other biopores or bulk soil. This already anticipated some important conclusions of the experiment: (1) the strong and fast impact of earthworms on OM dynamics and (2) the late stage of residue turnover in root biopores and their similarity to bulk soil after two years of decomposition. Principal component 1 clearly distinguished the 'fresh input' (earthworm influence; Fig. 1) from the 'old C' samples (no recent C input in bulk soil and root biopores). As earthworms were fed with clover grass during the entire experiment, they brought new C into the biopores and induced the 'earthworm effect'. The earthworm effect is most strongly correlated with GM/AX and (Ac/Al)_v ratios apparently indicative for low processed clover grass. Fig. 2 exemplarily illustrates (1) microbial processing, i.e. the red arrow shows the increase of the biomarker ratio from the undecomposed source biomass during biopore development, and (2) the earthworm effect, i.e. less decomposed earthworm biopore OM relative to bulk soil. Microbial processing includes processing in the earthworm gut by microflora, which is predominantly derived from soil microflora, and also enzymes produced by the earthworm itself (Curry and Schmidt, 2007). Earthworms replenished the OM, i.e. hemicelluloses (Fig. 2), lignin (V subunits, Fig. 3) and suberin and *n*-alkanes (Fig. 4) were less degraded than in bulk soil. This underlines the immense role of earthworms as ecosystem engineers (Jones et al., 1996).

The effect of the soil depth on both native earthworm and earthworm-incubated biopores was little: the degradation state was almost identical between soil depths and the variance of the proxies was smallest among the biopores (smallest ellipses in Fig. 1; Figs. 2–4). Therefore, even short-term earthworm burrowing for six months effectively added and redistributed OM within the burrows and equalised depth effects (Jégou et al., 2000). Earthworm incubations are almost independent of their duration beneficial for C and nutrient accumulation in the subsoil and could be easily implemented by reduced tillage (Athmann et al., 2017; Rumpel et al., 2012). While the soil depth had little effect on the OM in earthworm biopores, it was apparent for the

root biopores along PC 2 (Fig. 1). Root detritus was more decomposed in the lower subsoil (Fig. 2, Table 2C/N). The higher decomposition of root hemicelluloses (Fig. 2) may either result from a lower degree of lignification (Barros et al., 2015) or from a lower degree of cross-linking (arabinoxylan to lignin) of the younger root detritus in the deeper subsoil (Amin et al., 2014; Bertrand et al., 2005; Moorhead et al., 2014).

The root effect on all biomarker ratios was weaker than the earthworm effect (Figs. 2–4) and appeared to be wearing off after two years of root decomposition: only suberin and lignin (V subunits) were not yet as decomposed as the bulk SOM (Figs. 3 and 4). Root biopores had not yet fully reached the decomposition state, nor the C contents of the bulk SOM (Table 2). Thus, until less bioavailable inputs will be fully turned over at least 1–2 years more of accelerated C and nutrient turnover may be expected in the root biopores (Schöning et al., 2005).

4.3. OM processing in the biopores

The microbial processing (illustrated by the red and blue arrows in Figs. 2–4) in the biopore hotspots was lowest with the highest frequency of C input. First, taproot residues underwent the strongest microbial processing among biopores. Residues were probably only weakly incorporated into the surrounding soil as compared to earthworms' ingesting and casting (Lubbers et al., 2017). This limits the contact between root residues and mineral surfaces and, therefore, limits physicochemical C stabilisation (Schmidt et al., 2011). In contrast, intimate mixing of partly processed OM in the earthworm gut brings polar functional groups in contact with mineral surfaces and generates aggregates stabilised by organo-mineral interactions (Jégou et al., 2000; von Luetzow et al., 2006). It may be speculated that such protection might have contributed to the lower degree of processing of OM in earthworm biopores compared to root biopores (Figs. 2–4). For instance, the proxies of hemicelluloses and *n*-alkanes showed divergent trends with depth in root-influenced biopores (Figs. 2 and 4). The microbial processing of polysaccharides and lipids is, therefore, not governed by the same factors. Likely the varying interaction with the mineral phases due to oxidation state and functional groups as well as the low bioavailability of *n*-alkanes can help explain this behaviour (Rumpel et al., 2010, 2012; Vidal et al., 2016).

The frequent OM inputs by earthworms regularly replenished the decaying pool of easily bioavailable C (hemicelluloses) - fuelling a large-size microbial community of fast-growing Gram-negative bacteria, which preferentially grow on easily available C (Banfield et al., 2017; Curry and Schmidt, 2007; Hoang et al., 2016). The less bioavailable, structural OM (lignin, suberin) is then, consequently, only weakly decomposed (Figs. 3 and 4). In the case of the single OM input in the root biopores, a gradual conversion of plant biomass into more stabilised microbial necromass is likely (Puget and Le Drinkwater, 2001). The OM in root biopores enriched the microbial community in Gram-positive bacteria and actinobacteria (Banfield et al., 2017), i.e. slowly growing decomposers of structural and not easily available C

(Větrovský et al., 2014).

As hemicelluloses, suberin and lignin derive from primary and secondary plant cell walls and have similar functions and locations in plants (Kögel-Knabner et al., 2008), it was expected that they are decomposed at similar stages of decomposition (Cotrufo et al., 2013). However, the GM/AX ratio was more increased than the lignin oxidation state or HO-AA (relative to the inputs) demonstrating that the processing of hemicelluloses was faster than of lignin or suberin (Figs. 2–4). Therefore, the more easily available C of the hemicelluloses was preferentially processed.

OM processing in earthworm pores is dominated by two contrasting processes: (1) high microbial activity due to fresh C inputs likely caused intensive microbial processing of the biopores OM, and, (2) the biomarker ratios of earthworm biopores might be vastly dominated by the high frequency of imported ‘unprocessed’ OM from which only the highly bioavailable parts were utilised by microorganisms. Our observation of lower microbial processing especially of less bioavailable substance classes (lignin) points toward the dominance of the second process. The two processes of (1) increased microbial processing by high activity vs (2) dominance of frequent input of ‘unprocessed’ biomass can also be distinguished by the $\delta^{13}\text{C}$ value of the biopore OM. When earthworms accelerate the turnover of OM, increased fractionation processes take place leading to an increase of the $\delta^{13}\text{C}$ (Werth and Kuzyakov, 2010). In contrast, the imported fresh shoot litter (clover grass) decreased the $\delta^{13}\text{C}$ of biopore OM by the input of ‘unprocessed’ OM. This process is enhanced by the microbial preference for easily decomposable substance classes (Blagodatskaya et al., 2009), which are isotopically heavier. This results from processes during plant biomass synthesis (Hobbie and Werner, 2004; Park and Epstein, 1961) where lignin undergoes many fractionating reaction steps during its biosynthesis and becomes isotopically lighter than sugars which remain longer in the unreacted photosynthate pool and undergo fewer reaction steps to hemicelluloses (Boerjan et al., 2003; Pauly et al., 2013). The lower ^{13}C abundance of earthworm-incubated biopores (Table 2) confirms the relative accumulation of isotopically lighter substances (e.g. lignin) in biopores by the preferential decomposition of the isotopically heavier parts of the fresh clover grass litter as e.g. hemicelluloses (Figs. 2 and 3). Presumably, as soon as the C input ceases – e.g. in the root biopores – microorganisms would use all forms of C or even re-use, e.g. in the case of the bulk soil where the supply is very limited (Fierer et al., 2003; Richter and Markewitz, 1995), resulting in the higher $\delta^{13}\text{C}$ value of bulk SOM (Table 2).

The strong processing of bulk SOM indicated by its $\delta^{13}\text{C}$ value is also confirmed by highest (Ac/Al)_v ratios for bulk soil lignin. The bulk SOM derives from a wide range of inputs and was highly oxidised which is likely a pre-requisite for stabilisation of lignin (Vidal et al., 2016). The subunit-specific processing of lignin was similar in bulk soil and biopores: syringyl subunits were oxidised faster than vanillyl subunits (Heim and Schmidt, 2007; Thevenot et al., 2010). Although Banfield et al. (2017) found higher fungal/bacterial ratios in earthworm-incubated biopores, there was no indication that lignin processing in the biopore hotspots was mechanistically different from the bulk subsoil (Rasse et al., 2006). This may most likely be explained by the high functional redundancy of the microbial community in hotspots (Delmont et al., 2014) and the scarcity of specialists capable of co-metabolically attacking the lignin backbone (Tuomela et al., 2000; Větrovský et al., 2014). Suberin-derived lipids are also discussed as parts of the rather resistant SOM pool (Mendez-Millan et al., 2010). However, in bulk subsoil only very little intact suberin was recovered and the ratio of free depolymerisation products to suberin was highest (Fig. 4). Consequently, such long decomposition times as observed in bulk soil illustrated the intensive suberin decomposition of agricultural SOM and showed that intact suberin in the subsoil is mainly limited to hotspots.

4.4. The relevance of the biopore C turnover in the subsoil

Microbial hotspots are central to the understanding of subsoil C turnover (Kuzyakov and Blagodatskaya, 2015). Biopores are likely the largest subsoil hotspots by volume and almost all C and nutrient turnover in the subsoil is assumed to take place in hotspots (Kuzyakov and Blagodatskaya, 2015). Considering how long root biopores remained ‘hot’ (Figs. 2–4), they provide nutrients and C for time spans which are definitely long enough for subsequently grown crops to benefit from biopores’ nutrients. Due to their large number (e.g. 400 m⁻²; Athmann et al., 2013) and considerable C contents (Table 2), root biopores are very relevant for the total C input and SOM formation especially in the subsoil (Figs. 2–4) and therefore nutrient accumulation. Catch crop practices like greening incentivised by the EU’s Common Agricultural Policy could prove to have further long-term effects as root biopores are easily induced in great numbers by deep-rooting catch crops (Kautz et al., 2013; Wuest, 2001). Biopores can facilitate the acquisition of nutrients from the subsoil via (1) increasing the root-length density in the bulk soil or (2) uptake of nutrients from the biopore wall (Gaiser et al., 2012; Han et al., 2016; Jakobsen and Dexter, 1988; Kautz, 2015).

The earthworm biopores featured considerably higher C contents (Table 2), enzyme activities and higher nutrient mobilisation than the root biopores (Athmann et al., 2017; Banfield et al., 2017; Hoang et al., 2016), so the specific relevance of earthworm biopores for nutrient mobilisation is likely higher compared to root biopores. It remains unclear if earthworms overcompensate their forty times lower count (Dinter et al., 2013; Kautz et al., 2014) and actually cause higher C turnover than root biopores on the field scale. Being voids makes biopores attractive for new root growth which may actually hold back long-term C sequestration. Potentially stabilised OM in biopores, e.g. oxidised lignin (Fig. 3; Kiem and Kögel-Knabner, 2003; Vidal et al., 2016), may not be out of reach of roots: priming through biopore re-use makes direct C sequestration less likely (Kuzyakov, 2010). Consequently, biopores may mostly indirectly promote long-term C sequestration by increasing the total belowground C input by boosted nutrient cycling and subsoil access. If long-term C sequestration was mainly improved indirectly, then the earthworm biopores may support plant growth more than the slow-cycling root biopores.

5. Conclusions

Biopores are the largest-size hotspots of C and nutrient turnover in the subsoil, especially when induced by earthworms. The microbial processing of the C inputs was biopore-specific. More easily available parts of the OM (e.g. hemicellulose) were preferentially processed in biopores compared to lignin. The repeated inputs by earthworms relatively enrich less bioavailable and weakly processed OM (e.g. lignin) in the biopore environment. Earthworms incubation in root biopores bring new life to the root detritus: within six months they replenished the highly processed OM of the root biopores with only weakly processed shoot litter. Earthworms in root biopores led to 60% less processed hemicelluloses relative to root biopores. Consequently, even short periods of earthworm promotion have long-lasting effects on C contents and the degree of processing of the OM. Root necromass of perennials is likely to be processed roughly in the time it took to form the biomass (3–4 years). The nutrient accumulation and boosted mobilisation in root biopores hold up at least for durations relevant for crop rotations. Despite the 50% higher C accumulation in earthworm biopores, root biopores are up to forty times more frequent and therefore they quantitatively provide most of the OM input into subsoils. This may pose further incentives for growing perennial tap-rooted catch crops (also for shorter-term ‘greening’). We conclude that all biopore hotspots remain ‘hot’ at least for several years and, therefore, likely provide valuable nutrients along their preferential root growth pathways for subsequently grown crops. Thereby, biopores likely indirectly support long-term C sequestration through boosting plant productivity creating a

positive feedback on C sequestration.

Conflicts of interest

The authors declare no conflict of interest.

Funding

Role of the funding source: DFG KU 1184/29-1.

Acknowledgements

We highly acknowledge the support from the German Research Foundation (DFG KU1184/29 and DFG INST 186/1006-1). We would like to thank Timo Kautz and the colleagues from Institute of Organic Agriculture of the University of Bonn for establishing and managing the field trial Klein-Altendorf. We would also like to thank the Centre for Stable Isotope Research and Analysis Goettingen for $\delta^{13}\text{C}$ determinations.

Appendix A. Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.soilbio.2018.08.004>.

References

- Amblès, A., et al., 1994. *n*-Alkane oxidation in soil. Formation of internal monoalkenes. *Geoderma* 64, 111–124.
- Amelung, W., et al., 2008. Chapter 6 combining biomarker with stable isotope analyses for assessing the transformation and turnover of soil organic matter. In: In: Sparks, D.L. (Ed.), *Advances in Agronomy*, vol. 100. Elsevier/Academic Press, San Diego, pp. 155–250.
- Amin, B.A.Z., et al., 2014. Impact of fine litter chemistry on lignocellulolytic enzyme efficiency during decomposition of maize leaf and root in soil. *Biogeochemistry* 117, 169–183.
- Andriuzzi, W.S., Bolger, T., Schmidt, O., 2013. The drilosphere concept. Fine-scale incorporation of surface residue-derived N and C around natural *Lumbricus terrestris* burrows. *Soil Biology and Biochemistry* 64, 136–138.
- Angers, D.A., Mehuys, G.R., 1990. Barley and alfalfa cropping effects on carbohydrate contents of a clay soil and its size fractions. In: *Knowledge Gaps in Soil C and N Interactions*, vol. 22. pp. 285–288.
- Armas-Herrera, C.M., et al., 2016. Management effects on composition and dynamics of cutin and suberin in topsoil under agricultural use. *European Journal of Soil Science* 67 (4), 360–373.
- Athmann, M., et al., 2013. Root growth in biopores—evaluation with in situ endoscopy. *Plant and Soil* 371, 179–190.
- Athmann, M., et al., 2017. Six months of *L. terrestris* L. activity in root-formed biopores increases nutrient availability, microbial biomass and enzyme activity. *Applied Soil Ecology* 120, 135–142.
- Balesdent, J., Mariotti, A., Guillet, B., 1987. Natural ^{13}C abundance as a tracer for studies of soil organic matter dynamics. In: *Knowledge Gaps in Soil C and N Interactions*, vol. 19. pp. 25–30.
- Banfield, C.C., et al., 2017. Biopore history determines the microbial community composition in subsoil hotspots. *Biology and Fertility of Soils* 9, 54.
- Barros, J., et al., 2015. The cell biology of lignification in higher plants. *Annals of Botany* 115, 1053–1074.
- Bertrand, I., et al., 2005. Can the biochemical features and histology of wheat residues explain their decomposition in soil? *Plant and Soil* 281, 291–307.
- Blagodatskaya, E.V., et al., 2009. Contrasting effects of glucose, living roots and maize straw on microbial growth kinetics and substrate availability in soil. *European Journal of Soil Science* 60, 186–197.
- Boerjan, W., Ralph, J., Baucher, M., 2003. Lignin biosynthesis. *Annual Review of Plant Biology* 54, 519–546.
- Brown, G.G., Barois, I., Lavelle, P., 2000. Regulation of soil organic matter dynamics and microbial activity in the drilosphere and the role of interactions with other edaphic functional domains. *European Journal of Soil Biology* 36, 177–198.
- Bull, I.D., et al., 2000. Organic geochemical studies of soils from the Rothamsted classical experiments—V. The fate of lipids in different long-term experiments. *Organic Geochemistry* 31, 389–408.
- Butt, K.R., Lowe, C.N., 2007. A viable technique for tagging earthworms using visible implant elastomer. *Applied Soil Ecology* 35, 454–457.
- Chan, K., 2001. An overview of some tillage impacts on earthworm population abundance and diversity — implications for functioning in soils. *Soil and Tillage Research* 57, 179–191.
- Cheshire, M.V., Griffiths, B.S., 1989. The influence of earthworms and crane fly larvae on the decomposition of uniformly ^{14}C labelled plant material in soil. *Journal of Soil Science* 40, 117–124.
- Cotrufo, M.F., et al., 2013. The Microbial Efficiency-Matrix Stabilization (MEMS) framework integrates plant litter decomposition with soil organic matter stabilization: do labile plant inputs form stable soil organic matter? *Global Change Biology* 19, 988–995.
- Curry, J.P., Schmidt, O., 2007. The feeding ecology of earthworms – a review. *Pedobiologia* 50, 463–477.
- Delmont, T.O., et al., 2014. Microbial community development and unseen diversity recovery in inoculated sterile soil. *Biology and Fertility of Soils* 50, 1069–1076.
- Dignac, M.-F., et al., 2017. Increasing soil carbon storage. Mechanisms, effects of agricultural practices and proxies. A review. 37. *Agronomy for Sustainable Development*, pp. 351.
- Dinter, A., et al., 2013. Occurrence and distribution of earthworms in agricultural landscapes across Europe with regard to testing for responses to plant protection products. *Journal of Soils and Sediments* 13, 278–293.
- Eriksen-Hamel, N.S., et al., 2009. Earthworm populations and growth rates related to long-term crop residue and tillage management. *Soil and Tillage Research* 104, 311–316.
- Ertel, J.R., Hedges, J.I., 1984. The lignin component of humic substances. Distribution among soil and sedimentary humic, fulvic, and base-insoluble fractions. *Geochimica et Cosmochimica Acta* 48, 2065–2074.
- FAOSTAT, 2017. Agri-environmental Indicators—Land Use. <http://www.fao.org/faostat/en/#data/EL>, Accessed date: 3 January 2018.
- Fierer, N., Schimel, J.P., Holden, P.A., 2003. Variations in microbial community composition through two soil depth profiles. *Soil Biology and Biochemistry* 35, 167–176.
- Fontaine, S., et al., 2007. Stability of organic carbon in deep soil layers controlled by fresh carbon supply. *Nature* 450, 277–280.
- Frostegård, Å., Tunlid, A., Bååth, E., 2011. Use and misuse of PLFA measurements in soils. *Soil Biology and Biochemistry* 43, 1621–1625.
- Gaiser, T., et al., 2012. Evidence of improved water uptake from subsoil by spring wheat following lucerne in a temperate humid climate. *Field Crops Research* 126, 56–62.
- Glaser, B., Turrión, M.-B., Alef, K., 2004. Amino sugars and muramic acid—biomarkers for soil microbial community structure analysis. *Soil Biology and Biochemistry* 36, 399–407.
- Gunina, A., Kuzyakov, Y., 2014. Pathways of litter C by formation of aggregates and SOM density fractions: implications from ^{13}C natural abundance. *Soil Biology and Biochemistry* 71, 95–104.
- Gunina, A., Kuzyakov, Y., 2015. Sugars in soil and sweets for microorganisms: review of origin, content, composition and fate. *Soil Biology and Biochemistry* 90, 87–100.
- Haider, K., Martin, P., 1979. Abbau und Umwandlung von Pflanzenrückständen und ihren Inhaltsstoffen durch die Mikroflora des Bodens. *Zeitschrift für Pflanzenernährung und Bodenkunde* 142, 456–475.
- Han, E., Kautz, T., Köpke, U., 2016. Precrop root system determines root diameter of subsequent crop. *Biology and Fertility of Soils* 52, 113–118.
- Hedges, J.I., Ertel, J.R., 1982. Characterization of lignin by gas capillary chromatography of cupric oxide oxidation products. *Analytical Chemistry* 54, 174–178.
- Heim, A., Schmidt, M.W.I., 2007. Lignin turnover in arable soil and grassland analysed with two different labelling approaches. *European Journal of Soil Science* 58, 599–608.
- Hoang, D.T.T., et al., 2016. Hotspots of microbial activity induced by earthworm burrows, old root channels, and their combination in subsoil. *Biology and Fertility of Soils* 52, 1105–1119.
- Hobbie, E.A., Werner, R.A., 2004. Intramolecular, compound-specific, and bulk carbon isotope patterns in C 3 and C 4 plants. A review and synthesis. *New Phytologist* 161, 371–385.
- IPCC, 2014. Climate Change 2014: Synthesis Report. Contribution of Working Groups I, II and III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change. IPCC, Geneva, Switzerland.
- IUSS Working Group WRB, 2008. World reference base for soil resources 2006. Ein Rahmen für internationale Klassifikation, Korrelation und Kommunikation, first ed. BGR, Hannover.
- Jakobsen, B.E., Dexter, A.R., 1988. Influence of biopores on root growth, water uptake and grain yield of wheat (*Triticum aestivum*) based on predictions from a computer model. *Biology and Fertility of Soils* 6, 315–321.
- Jégou, D., et al., 2000. Burrowing activity of the earthworms *Lumbricus terrestris* and *Aporrectodea giardi* and consequences on C transfers in soil. *European Journal of Soil Biology* 36, 27–34.
- Jones, C., Lawton, J., Shachak, M., 1996. Organisms as ecosystem engineers. In: Samson, F.B., Knopf, F.L. (Eds.), *Ecosystem Management. Selected Readings*. Springer, New York, pp. 130–147.
- Kaiser, K., Kalbitz, K., 2012. Cycling downwards – dissolved organic matter in soils. *Soil Biology and Biochemistry* 52, 29–32.
- Kalbitz, K., Kaiser, K., 2008. Contribution of dissolved organic matter to carbon storage in forest mineral soils. *Journal of Plant Nutrition and Soil Science* 171, 52–60.
- Kautz, T., et al., 2013. Nutrient acquisition from arable subsoils in temperate climates: a review. *Soil Biology and Biochemistry* 57, 1003–1022.
- Kautz, T., et al., 2014. Contribution of anecic earthworms to biopore formation during cultivation of perennial ley crops. *Pedobiologia* 57, 47–52.
- Kautz, T., 2015. Research on subsoil biopores and their functions in organically managed soils: a review. *Renewable Agriculture and Food Systems* 30, 318–327.
- Kell, D.B., 2012. Large-scale sequestration of atmospheric carbon via plant roots in natural and agricultural ecosystems: why and how. *Philosophical Transactions of the Royal Society of London - B* 367, 1589–1597.
- Kiem, R., Kögel-Knabner, I., 2003. Contribution of lignin and polysaccharides to the refractory carbon pool in C-depleted arable soils. *Soil Biology and Biochemistry* 35, 101–118.

- Kindler, R., et al., 2011. Dissolved carbon leaching from soil is a crucial component of the net ecosystem carbon balance. *Global Change Biology* 17, 1167–1185.
- Kögel-Knabner, I., 2002. The macromolecular organic composition of plant and microbial residues as inputs to soil organic matter. *Soil Biology and Biochemistry* 34, 139–162.
- Kögel-Knabner, I., et al., 2008. Organo-mineral associations in temperate soils. Integrating biology, mineralogy, and organic matter chemistry. *Journal of Plant Nutrition and Soil Science* 171, 61–82.
- Kolattukudy, P.E., 1980. Biopolyester membranes of plants: cutin and suberin. *Science* 208, 990–1000.
- Kolattukudy, P.E., 1984. Biochemistry and function of cutin and suberin. *Canadian Journal of Botany* 62, 2918–2933.
- Kuzyakov, Y., 2010. Priming effects: interactions between living and dead organic matter. *Soil Biology and Biochemistry* 42, 1363–1371.
- Kuzyakov, Y., Blagodatskaya, E., 2015. Microbial hotspots and hot moments in soil: concept & review. *Soil Biology and Biochemistry* 83, 184–199.
- Lal, R., Negassa, W., Lorenz, K., 2015. Carbon sequestration in soil. *Current Opinion in Environmental Sustainability* 15, 79–86.
- Li, Y., et al., 2009. Climate change and drought: a risk assessment of crop-yield impacts. *Climate Research* 39, 31–46.
- Lichtfouse, E., et al., 1998. A novel pathway of soil organic matter formation by selective preservation of resistant straight-chain biopolymers: chemical and isotope evidence. *Organic Geochemistry* 28, 411–415.
- Lichtfouse, E., 1997. Heterogeneous turnover of molecular organic substances from crop soils as revealed by ^{13}C labeling at natural abundance with *Zea mays*. *Naturwissenschaften* 84, 23–25.
- Lubbers, I.M., Puleman, M.M., van Groenigen, J.W., 2017. Can earthworms simultaneously enhance decomposition and stabilization of plant residue carbon? *Soil Biology and Biochemistry* 105, 12–24.
- Mendez-Millan, M., et al., 2010. Molecular dynamics of shoot vs. root biomarkers in an agricultural soil estimated by natural abundance ^{13}C labelling. *Soil Biology and Biochemistry* 42, 169–177.
- Miltner, A., et al., 2012. SOM genesis: microbial biomass as a significant source. *Biogeochemistry* 111, 41–55.
- Moers, M., et al., 1990. Occurrence and origin of carbohydrates in peat samples from a red mangrove environment as reflected by abundances of neutral monosaccharides. *Geochimica et Cosmochimica Acta* 54, 2463–2472.
- Moorhead, D., et al., 2014. Interacting microbe and litter quality controls on litter decomposition. A modeling analysis. *PLoS One* 9, e108769.
- Munyakusi, E., et al., 1994. Earthworm macropores and preferential transport in a long-term manure applied typical hapludalf. *Journal of Environmental Quality* 23, 773.
- Naafs, D.F., van Bergen, P.F., 2002. A qualitative study on the chemical composition of ester-bound moieties in an acidic andosolic forest soil. *Organic Geochemistry* 33, 189–199.
- Oades, J.M., 1984. Soil organic matter and structural stability. Mechanisms and implications for management. *Plant and Soil* 76, 319–337.
- Otto, A., Simpson, M.J., 2007. Evaluation of CuO oxidation parameters for determining the source and stage of lignin degradation in soil. *Biogeochemistry* 80, 121–142.
- Otto, A., Simpson, M.J., 2007. Analysis of soil organic matter biomarkers by sequential chemical degradation and gas chromatography – mass spectrometry. *Journal of Separation Science* 30, 272–282.
- Pal, J.S., Giorgi, F., Bi, X., 2004. Consistency of recent European summer precipitation trends and extremes with future regional climate projections. *Geophysical Research Letters* 31.
- Park, R., Epstein, S., 1961. Metabolic fractionation of C_{13} & C_{12} in plants. *Plant Physiology* 36, 133.
- Pauly, M., et al., 2013. Hemicellulose biosynthesis. *Planta* 238, 627–642.
- Poeplau, C., Don, A., 2015. Carbon sequestration in agricultural soils via cultivation of cover crops – a meta-analysis. *Agriculture, Ecosystems & Environment* 200, 33–41.
- Puget, P., Le Drinkwater, 2001. Short-term dynamics of root- and shoot-derived carbon from a leguminous green manure. *Soil Science Society of America Journal* 65, 771–779.
- Quenea, K., et al., 2004. Variation in lipid relative abundance and composition among different particle size fractions of a forest soil. *Organic Geochemistry* 35, 1355–1370.
- Rasse, D.P., et al., 2006. Lignin turnover in an agricultural field: from plant residues to soil-protected fractions. *European Journal of Soil Science* 57, 530–538.
- Richter, D.D., Markewitz, D., 1995. How deep is soil? *BioScience* 600–609.
- Rumpel, C., Chabbi, A., Marschner, B., 2012. Carbon storage and sequestration in subsoil horizons: knowledge, gaps and potentials. In: Lal, R. (Ed.), *Recarbonization of the Biosphere. Ecosystems and the Global Carbon Cycle*. Springer Netherlands, Dordrecht, pp. 445–464.
- Rumpel, C., Eusterhues, K., Kögel-Knabner, I., 2010. Non-cellulosic neutral sugar contribution to mineral associated organic matter in top- and subsoil horizons of two acid forest soils. *Soil Biology and Biochemistry* 42, 379–382.
- Rumpel, C., Kögel-Knabner, I., 2011. Deep soil organic matter—a key but poorly understood component of terrestrial C cycle. *Plant and Soil* 338, 143–158.
- Rumpel, C., Kögel-Knabner, I., Bruhn, F., 2002. Vertical distribution, age, and chemical composition of organic carbon in two forest soils of different pedogenesis. *Organic Geochemistry* 33, 1131–1142.
- Scharlemann, J.P.W., et al., 2014. Global soil carbon. Understanding and managing the largest terrestrial carbon pool. *Carbon Management* 5, 81–91.
- Schimmel, D.S., 1995. Terrestrial ecosystems and the carbon cycle. *Global Change Biology* 1, 77–91.
- Schmidt, M.W.I., et al., 2011. Persistence of soil organic matter as an ecosystem property. *Nature* 478, 49–56.
- Schöning, I., Knicker, H., Kögel-Knabner, I., 2005. Intimate association between O/N-alkyl carbon and iron oxides in clay fractions of forest soils. *Organic Geochemistry* 36, 1378–1390.
- Simoneit, B.R.T., 2005. A review of current applications of mass spectrometry for biomarker/molecular tracer elucidations. *Mass Spectrometry Reviews* 24, 719–765.
- Six, J., et al., 2006. Bacterial and fungal contributions to carbon sequestration in agroecosystems. *Soil Science Society of America Journal* 70, 555–568.
- Spielvogel, S., Prielzel, J., Kögel-Knabner, I., 2007. Changes of lignin phenols and neutral sugars in different soil types of a high-elevation forest ecosystem 25 years after forest dieback. *Soil Biology and Biochemistry* 39, 655–668.
- Spielvogel, S., Prielzel, J., Kögel-Knabner, I., 2008. Soil organic matter stabilization in acidic forest soils is preferential and soil type-specific. *European Journal of Soil Science* 59, 674–692.
- Stirzaker, R.J., Passioura, J.B., Wilms, Y., 1996. Soil structure and plant growth: impact of bulk density and biopores. *Plant and Soil* 185, 151–162.
- Tanaka, H., et al., 1990. Determination of component sugars in soil organic matter by HPLC. *Zentralblatt für Mikrobiologie* 145, 621–628.
- Thakuria, D., et al., 2010. Gut wall bacteria of earthworms: a natural selection process. *The ISME Journal* 4, 357–366.
- Thevenot, M., Dignac, M.-F., Rumpel, C., 2010. Fate of lignins in soils. A review. *Soil Biology and Biochemistry* 42, 1200–1211.
- Tompkins, E.L., et al., 2010. Observed adaptation to climate change. UK evidence of transition to a well-adapting society. *Global Environmental Change* 20, 627–635.
- Torres-Sallan, G., et al., 2017. Clay illuviation provides a long-term sink for C sequestration in subsoils. *Scientific Reports* 7, 45635.
- Tuomela, M., et al., 2000. Biodegradation of lignin in a compost environment: a review. *Bioresource Technology* 72, 169–183.
- Urry, J., 2015. Climate change and society. In: Michie, J., Cooper, C.L. (Eds.), *Why the Social Sciences Matter*. Palgrave Macmillan UK, London, pp. 45–59.
- Větrovský, T., Steffen, K.T., Baldrian, P., 2014. Potential of cometabolic transformation of polysaccharides and lignin in lignocellulose by soil Actinobacteria. *PLoS One* 9, e89108.
- Vetterlein, D., et al., 2013. Illite transformation and potassium release upon changes in composition of the rhizosphere soil solution. *Plant and Soil* 371, 267–279.
- Vidal, A., et al., 2016. Molecular fate of root and shoot litter on incorporation and decomposition in earthworm casts. *Organic Geochemistry* 101, 1–10.
- von Luetzow, M., et al., 2006. Stabilization of organic matter in temperate soils. Mechanisms and their relevance under different soil conditions - a review. *European Journal of Soil Science* 57, 426–445.
- Werth, M., Kuzyakov, Y., 2010. ^{13}C fractionation at the root–microorganisms–soil interface: a review and outlook for partitioning studies. *Soil Biology and Biochemistry* 42, 1372–1384.
- Wiesenberg, G.L., et al., 2004. Source and turnover of organic matter in agricultural soils derived from *n*-alkane/*n*-carboxylic acid compositions and C-isotope signatures. In: *Advances in Organic Geochemistry 2003. Proceedings of the 21st International Meeting on Organic Geochemistry*, vol. 35. pp. 1371–1393.
- Wuest, S.B., 2001. Soil biopore estimation: effects of tillage, nitrogen, and photographic resolution. *Soil and Tillage Research* 62, 111–116.
- Zhang, X., Amelung, W., 1996. Gas chromatographic determination of muramic acid, glucosamine, mannosamine, and galactosamine in soils. *Soil Biology and Biochemistry* 28 (9), 1201–1206. [https://doi.org/10.1016/0038-0717\(96\)00117-4](https://doi.org/10.1016/0038-0717(96)00117-4).
- Zhang, H., Schrader, S., 1993. Earthworm effects on selected physical and chemical properties of soil aggregates. *Biology and Fertility of Soils* 15, 229–234.